

GraphRAG-Based Agent Architecture for Multimodal Knowledge Retrieval

Minor Project Report

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the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER ENGINEERING**



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DECLARATION

I, Arib Ansari (22BCS009), and Harshit Kumar Gupta (22BCS040), solemnly declare that the project work entitled "**GraphRAG-Based Agent Architecture for Multimodal Knowledge Retrieval**" is a record of our original work carried out under the supervision and guidance of Prof Sarfaraz Masood, Department of Computer Engineering, Jamia Millia Islamia, New Delhi. The work presented in this project is authentic, and any external sources utilized have been duly acknowledged.

We declare that this project has not been submitted previously in part or in full for the award of any degree or diploma in this institute or any other university. All the data, figures, and concepts taken from other sources have been appropriately cited and referenced.

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CERTIFICATE

This is to certify that the research work entitled “GraphRAG-Based Agent Architecture for Multimodal Knowledge Retrieval” conducted by **Arib Ansari** (Roll No.: 22BCS009) and **Harshit Kumar Gupta** (Roll No.: 22BCS040) is an authentic and original endeavour carried out under the mentorship and guidance in the Department of Computer Engineering at Jamia Millia Islamia, New Delhi.

The aforementioned project has been pursued diligently and ethically by the students as a part of their Bachelor of Engineering curriculum in Computer Engineering at Jamia Millia Islamia, during the academic year 2025-26. This project is presented as a culmination of their academic pursuits and to fulfil the requisites for the successful completion of their degree program.

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We would like to apologize for any unintentional omissions of names or contributions. The contributions of all involved are deeply appreciated and acknowledged.

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ABSTRACT

The exponential growth of multimodal data—spanning text, images, and audio—poses challenges for traditional information-retrieval systems, which remain largely text-centric despite advances in Retrieval-Augmented Generation (RAG). As a result, semantic signals in visual and acoustic data often remain underutilized, limiting AI systems’ ability to deliver fully context-aware, cross-modal responses.

This project presents a GraphRAG-based Multimodal Knowledge Retrieval Agent that unifies multimodal ingestion, structured storage, and graph-native retrieval. Text is embedded and indexed, images are analyzed using vision-language models to extract objects and relationships, before all entities and relations are encoded as nodes and edges in a provenance-rich knowledge graph.

Using GraphRAG’s hierarchical community detection, the system organizes related entities into nested clusters and performs hybrid retrieval via global summary traversal followed by local k-hop exploration. Retrieved multimodal context is synthesized into cohesive, evidence-backed responses capable of referencing source text, and images enabling complex reasoning across a unified multimodal knowledge space.

Keywords: GraphRAG, Multimodal Retrieval, Knowledge Graphs, Chart Reasoning, YOLO Segmentation, Neo4j, Cross-Modal Fusion, RAG Evaluation

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CMEL	Cross-Modal Entity Linking
GPU	Graphics Processing Unit
KG	Knowledge Graph
LLaVA	Large Language-and-Vision Assistant
LLM	Large Language Model
MLLM	Multimodal Large Language Model
MMKG	Multimodal Knowledge Graph
MMGraphRAG	Multimodal Graph Retrieval-Augmented Generation
NER	Named Entity Recognition
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
SOTA	State-of-the-Art
VRAM	Video Random Access Memory
YOLO	You Only Look Once (Object Detection Model)

1.INTRODUCTION

1.1 Background and Motivation

The digital landscape has fundamentally transformed how we create, store, and access information. Today, knowledge exists in increasingly diverse forms—text documents, scientific papers, business presentations filled with charts and graphs, instructional videos with spoken narration, and annotated diagrams. Yet, despite this multimodal richness in our data sources, most artificial intelligence systems still operate as though information exists only in text form.

Large Language Models (LLMs) have revolutionized natural language understanding and generation, demonstrating remarkable capabilities across countless tasks. However, like traditional search engines, they remain fundamentally limited: they can only reason about text, and when we try to force them to understand images or audio, we must first convert those modalities into crude text descriptions, losing much of their original context and precision.^[8]

This gap between how information actually exists in the world—richly integrated across multiple modalities—and how our AI systems process it represents a critical inefficiency in modern knowledge work. Consider a student studying from textbooks with charts, diagrams, and illustrations; a researcher reviewing literature with figures and data tables; or a professional synthesizing technical documentation with visual elements. All of them perform what humans do naturally: they integrate information across different forms to build comprehensive understanding. Yet our AI assistants cannot replicate this integrated reasoning, forcing them to sacrifice accuracy and completeness.

The motivation for this project is straightforward: extend RAG's integration capabilities to multimodal information, enabling AI systems to reason coherently across text and images within a single unified framework.

1.2 GraphRAG and the Need for Multimodal Fusion

1.2.1 Evolution of RAG Systems

Retrieval-Augmented Generation (RAG) grounds large language models in external knowledge by retrieving relevant information at query time.^[8] This approach improves factual accuracy and reduces hallucinations, making RAG the standard architecture for knowledge-intensive applications. However, conventional “vector RAG” treats all information as flat text chunks and retrieves them solely through embedding similarity. As a result, it fails to capture semantic structure, entity relationships, or hierarchical organization within the data. While adequate for

localized queries, vector RAG performs poorly when tasks require multi-hop reasoning, evidence synthesis from multiple documents, or navigation across conceptual hierarchies.

Moreover, traditional RAG is inherently text-centric and cannot fully incorporate images, charts, or diagrams. Visual formats encode meaning through spatial layout, numerical precision, and patterns that cannot be preserved through textual representation alone. This limitation is reflected in empirical benchmarks, where multimodal models perform significantly worse on chart-based question answering than on text-only tasks, underscoring the persistent difficulty of visual reasoning within RAG systems.^[20]

1.2.2 GraphRAG: Structural Reasoning Through Knowledge Graphs

GraphRAG shifts retrieval from vector similarity to explicit structural reasoning by constructing semantically rich knowledge graphs.^[7] Entities become nodes, relationships are represented as labeled edges, and hierarchical community detection (e.g., Leiden) groups related entities^[18] into nested clusters with summary nodes at multiple abstraction levels. This representation preserves semantic context, relational structure, and multi-scale organization.

At query time, GraphRAG uses a two-stage retrieval process^[7]:

1. Global retrieval identifies relevant high-level communities using summaries.
2. Local retrieval performs k-hop traversal within selected communities to surface fine-grained evidence.

Unlike vector RAG, GraphRAG supports multi-hop reasoning, relational inference, and evidence aggregation across distributed information sources.^[9] The hierarchical abstraction enables efficient navigation from broad topic areas to precise contextual details, making GraphRAG substantially more effective for complex queries requiring structured, interpretable reasoning.

1.3 Project Scope and System Overview

This project develops a GraphRAG-based Multimodal Knowledge Retrieval Agent integrating text and images through six stages:

1. Multimodal Ingestion: YOLOv8^[19] segmentation, LLaVA^[10] descriptions, and scene graphs from images; tokenization and NER from text
2. Cross-Modal Entity Extraction: Normalize entities from both modalities with explicit modal provenance
3. Neo4j Graph Construction: Store entities as nodes with cross-modal links between text and image elements^[26]
4. Hierarchical Community Detection: Use Leiden algorithm^[18] to organize entities into nested clusters preserving modal information

5. Hybrid Two-Stage Retrieval: Global community search + local k-hop traversal for efficient cross-modal reasoning, retrieving relevant multimodal context with full source attribution
6. Generation and Personalization: Retrieved context is synthesized into evidence-backed answers through an LLM. The system maintains personalized user models, remembering which entities and documents have been useful in past conversations and using this memory to improve future retrievals.

1.4 Objectives and Key Contributions

- **End-to-End Multimodal GraphRAG System.**
We implement a complete multimodal GraphRAG pipeline—spanning ingestion, graph construction, retrieval, and answer generation—demonstrating that multimodal, graph-based retrieval is practically realizable beyond theoretical proposals.
- **Empirical Evaluation on Chart-Based QA.**
Using the Chart-MRAG benchmark (4,738 QA pairs), our system outperforms embedding-only baselines on information-dense visual queries, addressing a major gap in multimodal RAG evaluation.
- **Cross-Modal Knowledge Integration.**
The system unifies textual and visual entities through structured alignment and fusion, enabling genuine multimodal reasoning rather than text-only approximations of visual information.
- **Personalized Graph Memory.**
We introduce a user-specific knowledge graph with long-term memory, allowing retrieval and generation to adapt to individual user histories, thereby overcoming the generic behavior of conventional RAG systems.
- **Insights into Multimodal Reasoning.**
Our findings highlight that:
 - (a) perfect retrieval does not guarantee accurate generation due to inherent LLM limitations;
 - (b) chart reasoning differs fundamentally from text reasoning; and
 - (c) robust cross-modal alignment is essential for effective multimodal inference.

2.LITERATURE REVIEW

2.1 Retrieval-Augmented Reasoning: Evolution Toward Structured and Multimodal Intelligence

Recent advances in large language models (LLMs) have significantly accelerated research on retrieval-augmented systems capable of incorporating external knowledge to enhance reasoning accuracy. Early Retrieval-Augmented Generation (RAG) methods centered on embedding-based text retrieval to supplement model outputs, enabling improved factual grounding in knowledge-intensive tasks.^[8] However, the rapidly expanding diversity of data modalities—text, images, charts, tables, and diagrams—prompted the development of more structured retrieval approaches. Contemporary research now emphasizes graph-based representations, hierarchical retrieval strategies, and multimodal knowledge integration to support deep reasoning, global sensemaking, and cross-modal alignment. This evolution lays the foundation for modern frameworks such as GraphRAG, Multimodal GraphRAG, and multimodal knowledge graph-based reasoning systems.

2.2 Graph-Based Retrieval: Knowledge Graphs as Structured Indices

Graph-based retrieval models emerged as a solution to the representational limitations of flat vector databases. By structuring knowledge as entities and relationships, knowledge graphs (KGs) enable multi-hop traversal, semantic clustering, and global contextual understanding across large corpora.^[24] These graph-oriented methods support fine-grained reasoning pathways absent in conventional embedding-based retrieval, thereby enabling LLMs to generate more coherent, evidence-grounded outputs.

2.2.1 GraphRAG: Hierarchical Knowledge Graph-Driven Retrieval

GraphRAG represents a major breakthrough in retrieval augmentation by transforming unstructured corpora into structured semantic graphs. Edge et al.^[7] demonstrate how entity extraction, graph construction, and community detection methods such as Leiden clustering^[18] facilitate hierarchical summarization and multi-hop reasoning across entire document collections. GraphRAG’s ability to retrieve subgraphs based on thematic clusters or related entity chains enables deep contextual integration well beyond traditional chunk-level retrieval.

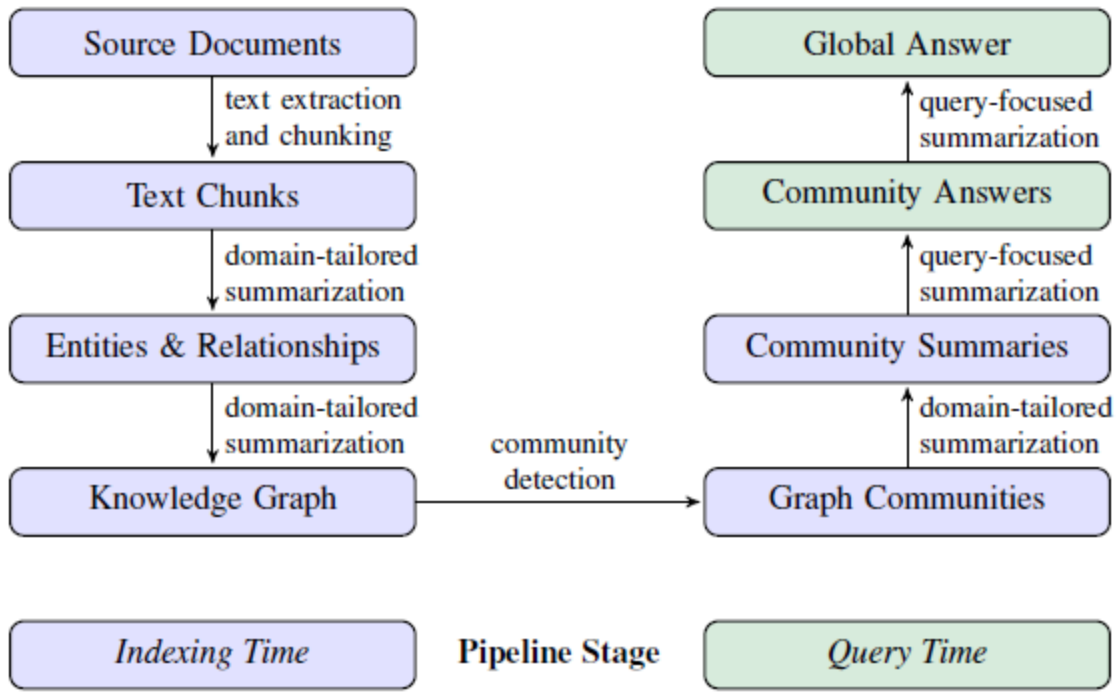


Figure 2.1: Graph RAG pipeline using an LLM-derived graph index of source document text.^[7]

2.2.2 MMGraphRAG: Multimodal Graph-Based Retrieval-Augmented Generation

The MMGraphRAG framework represents a major advancement by extending GraphRAG into multimodal domains. Unlike text-only GraphRAG, MMGraphRAG constructs a Multimodal Knowledge Graph (MMKG) by extracting entities and relationships from text, images, diagrams, tables, and charts.^[20]

- Cross-Modal Entity Linking (CMEL) to map visual elements (e.g., chart labels, plot regions, diagram symbols) to textual entities.
- Spectral clustering to group semantically related multimodal components.
- Multimodal fusion via graph message passing to combine textual, visual, and structural cues.
- Cross-modal retrieval enabling the system to answer queries involving images, plots, or both.

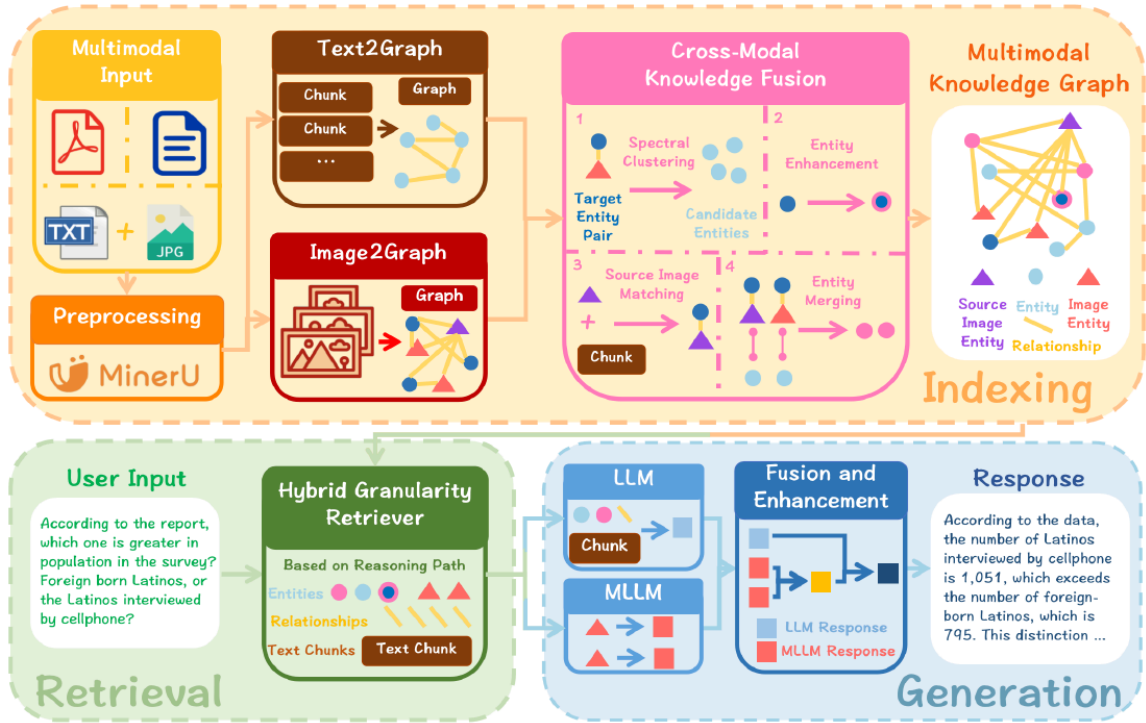


Figure 2.2: MMGraphRAG Framework Overview^[20]

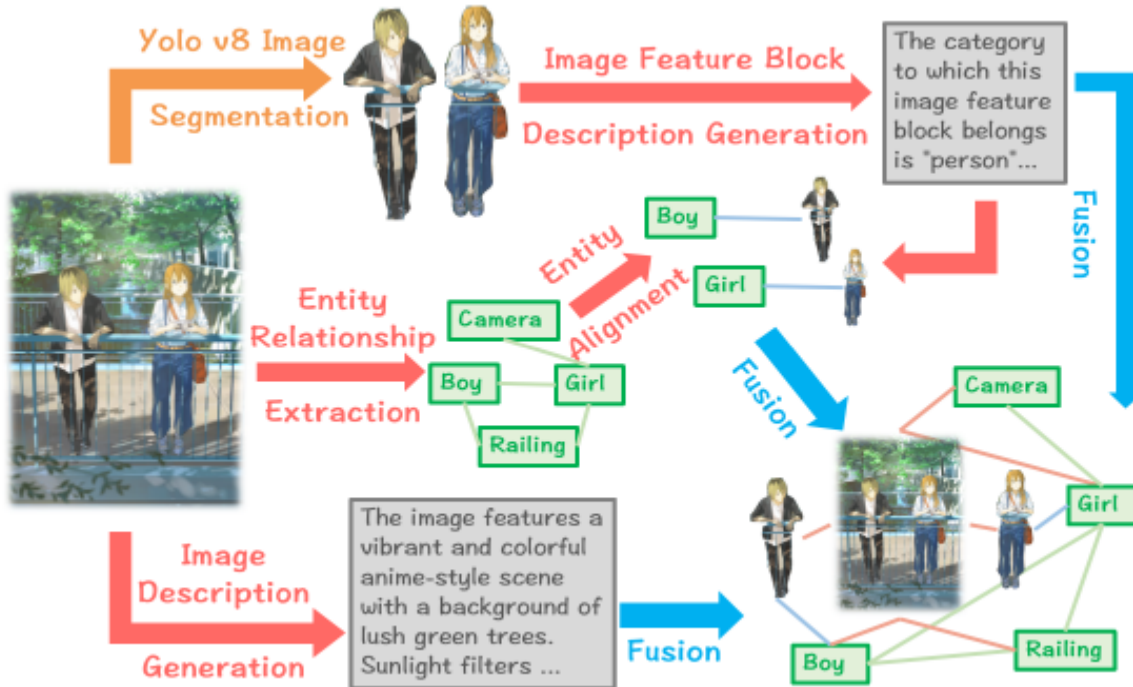


Figure 2.3: An Example of the Img2Graph Module in Action^[20]

2.3 Multimodal Knowledge Graphs and Performance in Complex Benchmarks

Modern multimodal reasoning systems are evaluated using datasets requiring visual–textual integration.

2.3.1 ScienceQA Benchmark

Benchmarking studies for multimodal reasoning systems increasingly rely on datasets that evaluate both visual–textual understanding and structured retrieval capabilities. A principal benchmark in this space is **ScienceQA**^[25], which assesses a model’s ability to integrate diagrams, images, and textual explanations to answer multi-step scientific questions. Studies incorporating graph-based retrieval, such as MGraphRAG^[13], report improved performance on ScienceQA due to the use of multimodal knowledge graphs that provide entity-level grounding and clearer relational pathways between visual and textual elements. This structured representation enables more accurate reasoning compared to conventional vision–language pipelines.

2.3.2 Chart-Based Benchmarks (Chart-MRAG)

Complementing ScienceQA, the **Chart-MRAG**^[28] benchmark focuses specifically on chart-based question answering, emphasizing numerical understanding, OCR alignment, and reasoning over plotted data. Chart-MRAG evaluates whether systems can reliably interpret chart components—such as axes, labels, and data points—while maintaining factual consistency. MMGraphRAG demonstrates meaningful improvements on Chart-MRAG by transforming chart elements into graph structures that preserve quantitative relationships and visual semantics. This graph-oriented approach allows retrieval to operate over structured chart entities rather than raw pixels or unstructured text, enabling more precise and interpretable chart reasoning.

Comparative Assessment of Retrieval, Graph-Based, and Multimodal Approaches ^{[7],[8],[13],[20],[24],[28]}

Table 2.1 — Comparison of Recent Approaches in Retrieval-Augmented, Graph-Based, and Multimodal Reasoning

Authors	Paper Name & Publisher	Proposed Approach	Remarks
Gao et al. (2024)	Retrieval-Augmented Generation Survey	Presents a structured taxonomy of RAG paradigms, including Naive, Advanced, and Modular RAG; outlines core retrieval-generation pipelines.	Establishes foundational principles of RAG; highlights evaluation benchmarks such as ScienceQA and multimodal QA tasks; frames limitations addressed by later graph-based methods.
Edge et al. (2024)	GraphRAG : Enhancing RAG with Knowledge Graphs	Constructs entity-based knowledge graphs from text; uses community detection and hierarchical summaries for retrieval.	Enables global sensemaking and multi-hop reasoning; outperforms chunk-based RAG in large corpus synthesis tasks.
Zhang et al. (2025)	GraphRAG Survey	Categorizes graph-based retrieval models into knowledge-based, index-based, and hybrid approaches; analyzes system design choices.	Highlights advantages of graph-structured retrieval for semantic coherence and corpus-scale contextual reasoning.
Xueyao Wan, Hang Yu (2025)	MMGraphRAG: Multimodal Graph Retrieval-Augmented Generation	Builds a Multimodal KG using cross-modal entity linking (CMEL) and spectral clustering to align visual-textual concepts.	Demonstrates improved performance on ScienceQA and multimodal QA tasks; shows strong cross-modal grounding and retrieval precision.
Lee et al. (2024)	MR-MKG: Multimodal Reasoning via Multimodal Knowledge Graphs	Integrates scene graphs and textual triples using GNN-based fusion for multimodal QA.	Yields accuracy improvements and reduces hallucinations; emphasizes interpretable reasoning paths.
Yuming Yang et al. (2025)	Chart-MRAG: Multimodal Graph Construction for Chart Reasoning	Converts chart elements (axes, labels, numeric points) into graph nodes and relationships for structured retrieval.	Enhances numeric reasoning and factual consistency in chart-based QA; well-suited for tasks requiring OCR and quantitative interpretation.

2.4 Conclusion

The reviewed literature presents a clear developmental trajectory from early text-only retrieval systems to sophisticated graph-based and multimodal reasoning frameworks. GraphRAG introduced structured entity-centric retrieval and global sensemaking, while MMGraphRAG extended these capabilities into multimodal domains through unified knowledge graph construction and cross-modal alignment. Benchmark results on datasets such as ScienceQA and ChartQA demonstrate the effectiveness of graph-based multimodal retrieval for tasks requiring integrated visual and textual reasoning. These contributions collectively justify the methodological approach adopted in the present study.

3. Proposed Methodology

This project implements a Multimodal GraphRAG agent that constructs, enriches, and queries a personalized multimodal knowledge graph in Neo4j, then evaluates the system on ScienceQA^[25] and Chart-MRAG^[28]. The method is organized into four stages: (1) Neo4j schema & graph building; (2) multimodal ingestion, graph enrichment and cross-modal fusion; (3) retrieval with community detection, k-hop traversal and memory weighting; and (4) benchmarking and evaluation.

3.1 Graph Schema

The system is built on a structured multimodal knowledge graph in Neo4j^[26], which serves as the core data backbone for all ingestion, fusion, and retrieval processes. The schema used (as shown in the accompanying tables and schema diagram) defines a compact yet expressive set of node types—**Entity**, **Chunk**, **Image**, **User**, **Community**, **Query**, and **Answer**—each capturing a different aspect of the user’s multimodal knowledge space. Entities store semantic concepts extracted from text or visual descriptions, while Chunk and Image nodes preserve the original evidence sources. User nodes anchor personalization and isolation of each user’s subgraph, and Community nodes later support clustering for global contextual retrieval. Query and Answer nodes enable long-term Q/A memory.

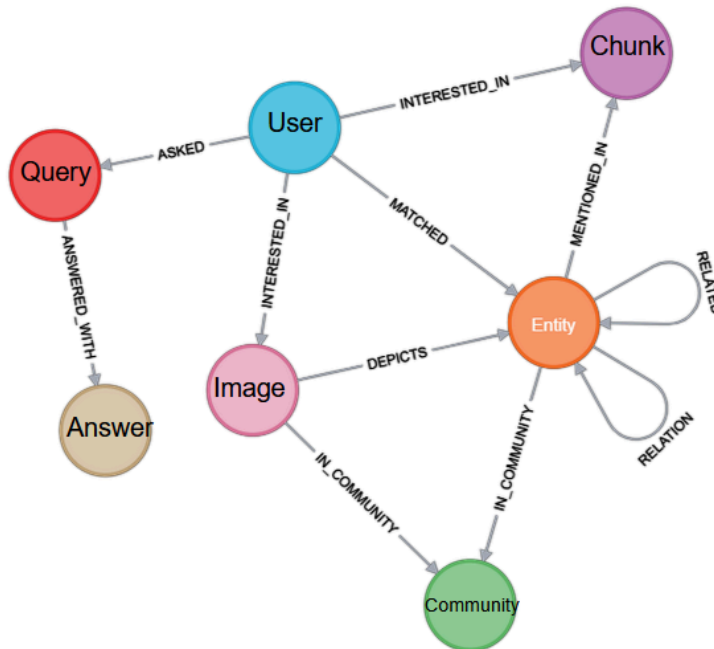


Figure 3.1: Neo4j Schema Visualization

Relationships such as **MENTIONED_IN**, **DEPICTS**, **RELATED**, and **RELATION** encode textual and visual links, whereas **INTERESTED_IN**, **ASKED**, and **ANSWERED_WITH** capture personalization, history, and system interactions. Confidence scores, timestamps, and provenance metadata attached to these edges allow downstream retrieval algorithms to apply weighted scoring and relevance ranking.

Table 3.1: Relationship Types

Relationship Type	Link	Purpose in RAG
INTERESTED_IN	User -> Chunk or Image	Defines the user's personal subgraph and ingested sources.
MENTIONED_IN	Entity -> Chunk	Links entities to the text chunk they came from.
DEPICTS	Image -> Entity	Links images to the visual entities they contain (Cross-Modal Link).
RELATION	Entity - Entity	Explicit structural relations extracted from the content.
RELATED	Entity - Entity	Enriched or inferred relations (e.g., from the LLM enrichment step).
IN_COMMUNITY	Entity or Image -> Community	Groups nodes into clusters following the Leiden algorithm.
ASKED	User -> Query	Connects user to their question (Q/A memory).
ANSWERED_WITH	Query -> Answer	Connects question to the generated answer (Q/A memory).
MATCHED	User -> Entity	Positive Reinforcement for successfully used entities in retrieval.

Neo4j is initialized with indexes and constraints on high-frequency properties (e.g., **Entity.name**, **Chunk.id**) to ensure efficient lookups during ingestion and retrieval. A persistent Bolt driver connection enables batch insertion of nodes and relationships as the system processes text, PDFs, images, and user queries. Each ingested element is assigned modality metadata and its original source, ensuring traceability and supporting later fusion stages.

Table 3.2: Node Labels

Label	Purpose in RAG
Entity	The core concepts, people, and objects (from text or images).
Chunk	Text segments from ingested documents/text.
Image	Nodes for uploaded images/visual context.
User	The active user for personalization and subgraph scoping.
Community	Clusters of related nodes used for summarization.
Query	Stores the user's question as part of Q/A Memory.
Answer	Stores the system's response as part of Q/A Memory.

3.2 Multimodal Ingestion

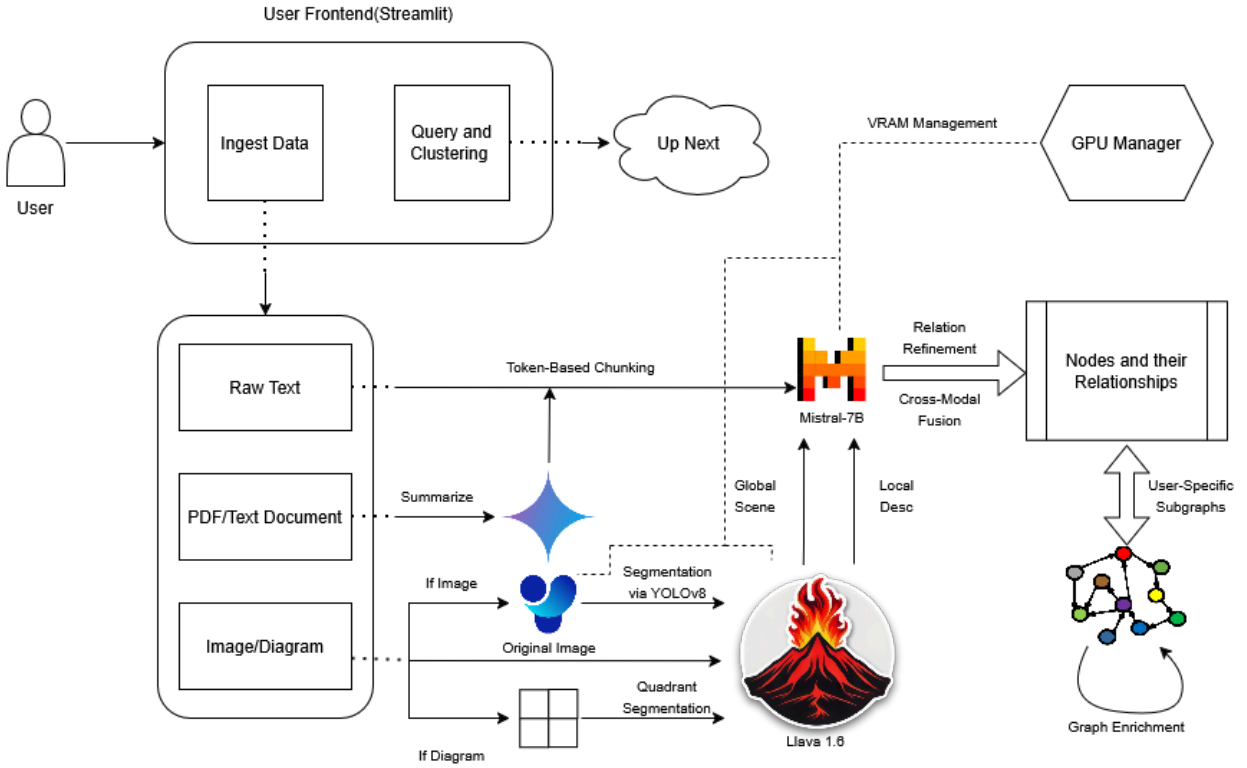


Figure 3.2 : KG Creation Pipeline

3.2.1 Text Chunking and PDF Summarization

Text inputs are first normalized and segmented so they can be reliably processed by local and cloud-based LLMs. The system performs:

- **Token-based chunking** of long documents with overlap to preserve context.
- **Cleaning & normalization** to ensure consistency across sources.
- **Gemini^[2] PDF summarization**, which extracts high-quality text, section boundaries, and resolves layout artifacts such as multi-column flow or figure interruptions.
- **Local Mistral-7B^[27] processing**, which identifies entities and proposes initial relations for each chunk.

This two-step Gemini → Mistral workflow produces stable Chunk nodes and candidate Entity nodes that form the textual basis of the knowledge graph.

3.2.2 Visual Scene Graph Generation

Images, diagrams, and charts are converted into structured semantic representations using a dedicated vision pipeline:

- **YOLOv8^[19] detection:** localizes objects, diagram components, axis labels, and regions of interest.
- **Quadrant segmentation (if needed):** improves precision on dense or cluttered diagrams.
- **LLaVA 1.6^[10] captioning & scene graph extraction:** generates
 - entity descriptions,
 - attribute triples,
 - simple relational triples.

The Global and Local scenes extracted by LLaVa then get sent into mistral for entity-relation extraction. The output is stored as temporary **VisualEntity** nodes linked to corresponding **Image** nodes, creating a structured representation of visual information.

3.2.3 Semantic Alignment and Reasoning

To avoid redundancy and ensure conceptual consistency, the system attempts to align each VisualEntity with existing textual entities. A candidate search retrieves potentially matching textual entities using fuzzy matching^[3] and optional embeddings^[1]. These candidates are then evaluated using the local Mistral-7B reasoning model, which determines whether the visual and textual descriptions refer to the same underlying concept. Only high-confidence matches proceed.

Following entity-level matching, the system performs a second Mistral reasoning pass for relation inference. Here, the model examines descriptions, surrounding context, and nearby nodes in the graph to propose additional edges that may be missing. These suggestions help strengthen semantic structure before more advanced graph operations occur.

3.2.4 Graph Fusion and Subgraph Reinforcement

Once alignment is confirmed:

- The **VisualEntity** is merged into the corresponding **Entity** node.
- The resulting node becomes a **Multimodal Entity**, combining text-based and visual-based information.
- Visual provenance is retained via **DEPICTS** edges.
- The fused entity becomes more prominent in retrieval due to increased connectivity.

This reinforcement ensures that whether a user asks through text, shows an image, or queries relationally, the system resolves to the same conceptual node.

3.2.5 GPU Manager for Efficient Model Execution

The ingestion pipeline relies on multiple computationally intensive models—YOLOv8, LLaVA, and Mistral-7B—each requiring significant GPU memory. To ensure stable and efficient

operation, the system uses a dedicated GPU Manager that loads models only when needed, executes their tasks, and then unloads them immediately. This prevents VRAM exhaustion, avoids model conflicts, and enables ingestion to scale across long documents and many images. The GPU Manager also provides predictable latency by serializing vision and language workloads, making multimodal processing both reliable and resource-efficient.

3.2.6 Graph Enrichment Through Orphan–Anchor Linking

After ingestion and fusion, the graph contains both well-connected concepts and newly introduced entities with minimal context. Enrichment strengthens graph structure by linking these together.

Identifying Node Groups

- **Orphans:**
 - Weakly connected or newly created nodes
 - Typically come from single mentions or small visual regions
- **Anchors:**
 - Highly connected nodes frequently referenced throughout the data
 - Represent stable concepts in the user's knowledge graph

Enrichment Process

- A small subset of orphans and anchors is selected.
- The system prompts **Mistral-7B** with context about both groups.
- The LLM proposes meaningful relations (e.g., functional, categorical, associative).
- Only coherent, non-duplicate, and contextually valid relations are added.

3.3 Retrieval and Clustering

The retrieval pipeline forms the core of the system’s interactive question-answering capability. It converts a user query into structured search operations over the personalized multimodal knowledge graph, gathers relevant evidence, and synthesizes a final answer using Gemini. Retrieval is combined with community detection via the Leiden algorithm^[18], allowing the system to reason not only over isolated facts but also across topically coherent clusters of information.

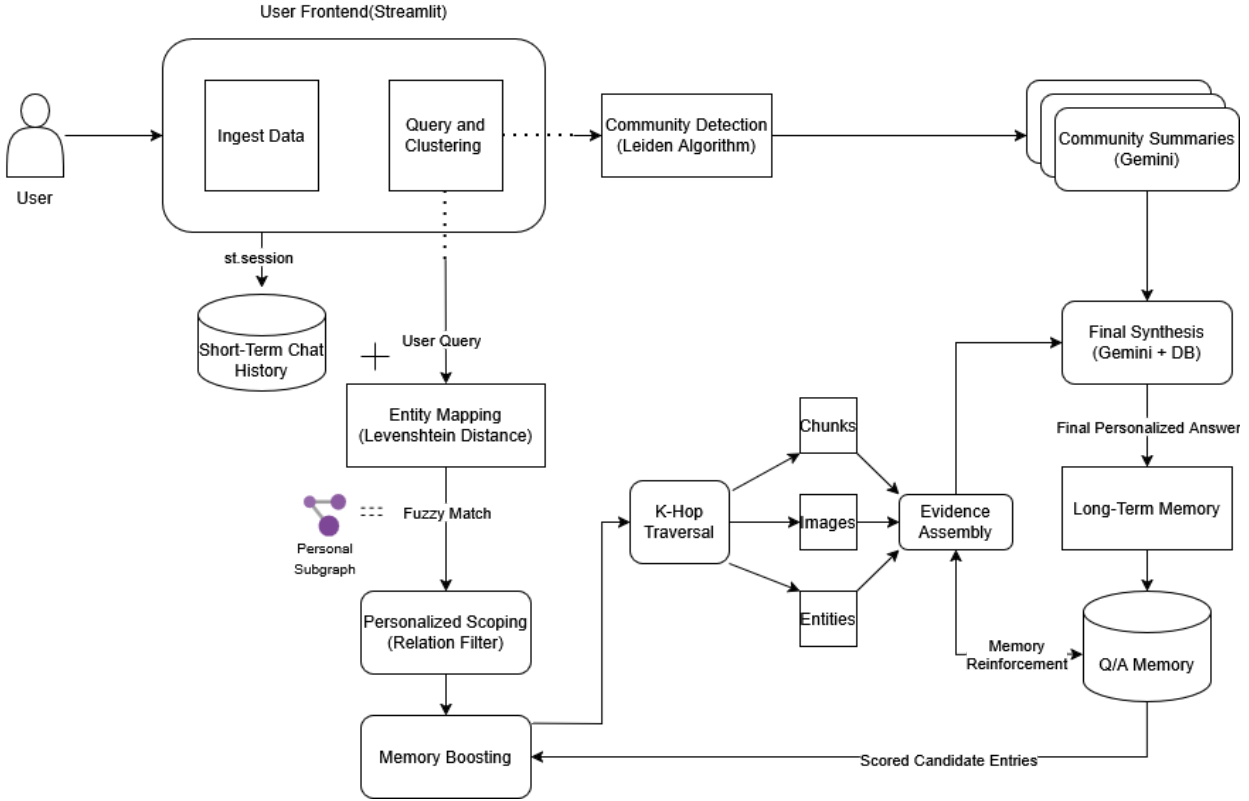


Figure 3.3: Query Synthesis and Response Generation

3.3.1 Query Interpretation and Entity Mapping

When a query is received, the system first interprets its semantic components and maps them to candidate entities in the user’s graph. The process begins with lightweight text analysis and continues through a multi-stage matching logic:

- **Keyword and phrase extraction** isolates important semantic units.
- **Levenshtein-based^[22] fuzzy matching^[3]** evaluates how closely each graph entity resembles the query’s terms.
- **Overlap scoring and substring matching** help identify partial or approximate matches when the query is less explicit.
- **Strict personal-subgraph filtering** ensures that only entities connected to the user (via INTERESTED_IN or CREATED edges) are considered.

3.3.2 Memory-Boosted Ranking

Retrieval is personalized by incorporating short-term and long-term memory signals. Short-term memory tracks the recent conversation window, allowing the system to infer when the user is

continuing a topic. Long-term memory stores broader summaries of recurring interests, allowing stable preference patterns to influence retrieval.

These memory sources affect ranking through:

- **Interest reinforcement** — entities referenced successfully in previous answers receive a small, cumulative score increase.
- **Session continuity** — recent topics are automatically weighted higher.
- **User familiarity modeling** — nodes frequently interacted with become more prominent.

3.3.3 K-Hop Retrieval and Evidence Assembly

A controlled **k-hop traversal** (2 hops for us) expands outward from the selected entities. Only meaningful semantic edges—primarily *RELATION* and *RELATED*—are followed. This produces a compact set of chunks, images and related entities.

The resulting material is converted into concise evidence snippets, for example:

- [TEXT cid] ...relevant excerpt...
- [IMAGE cid] summary
- [ENTITY cid] brief definition

Together, these form the factual grounding for the final answer.

3.3.4 Community Detection and Clustering

Beyond isolated nodes, the system also reasons across broader themes within the knowledge graph. This is accomplished through **community detection**, implemented using the Leiden algorithm^[18].

The algorithm groups entities, chunks, and images into topically coherent clusters based on graph connectivity. These clusters:

- capture the user’s areas of interest,
- assist in resolving broad or ambiguous queries, and
- provide structured summaries of large domains.

When the query aligns with a community, the system retrieves that community’s summary—generated earlier using Gemini—and incorporates it into the reasoning pipeline.

This allows the system to answer not only factual questions but also conceptual or thematic ones.

3.3.5 Personalized Scoping and Fuzzy Expansion

To maintain precision while still supporting explorative reasoning, retrieval operates within a personalized scope.

- **Relation filters** ensure expansion only occurs across edges meaningful to the user’s graph.
- **Fuzzy^[3] expansion** allows the system to include near-matches when the query lacks exact terminology, as long as they remain semantically relevant.
- **Scope narrowing** prevents unrelated branches of the graph from influencing retrieval.

3.3.6 Final Answer Synthesis with Gemini

The final stage transforms structured retrieval outputs into a coherent natural-language response.

Gemini receives the original user query. Assembled evidence, relevant communities(if applicable) and memory context.

Using these inputs, it synthesizes a grounded, context-aware answer that reflects the user’s personal knowledge graph rather than generic world knowledge.^[2]

3.4 Benchmarking Framework

To evaluate the performance of the proposed multimodal GraphRAG system, we adopt a dual-benchmark strategy combining **ScienceQA**^[25]—which emphasizes multimodal scientific reasoning—and **Chart-MRAG**^[28]—which assesses retrieval and reasoning over dense, structured charts. Together, these datasets stress-test our system across both *semantic multimodal reasoning* and *layout-heavy numerically grounded reasoning*.

Our benchmarking scripts ([benchmark.py](#), [benchmark_sqa.py](#)) implement a unified evaluation pipeline that feeds queries through the full retrieval stack—entity mapping, k-hop traversal, evidence assembly, and LLM synthesis—mirroring real user interactions.

3.4.1 ScienceQA Evaluation

ScienceQA is a multimodal dataset containing diagrams, short textual descriptions, and scientific reasoning questions.^[25] Each example is first ingested into the knowledge graph through the standard pipeline:

- Text is chunked and entities are extracted using Mistral-7B,
- Diagrams/Images are processed through YOLOv8 + LLaVA-1.6 to form scene-graph triples,
- Visual and textual representations are fused into unified multimodal entities.

Once the ScienceQA subgraph is built, each question is executed like a normal user query. The system performs entity matching, expands relevant nodes through k-hop traversal, assembles text and visual evidence, and finally synthesizes an answer using Gemini.

This benchmark assesses the system’s ability to retrieve grounded information from both text and diagram components, align multimodal entities, and generate logically coherent, evidence-based answers.

3.4.2 Chart-MRAG Evaluation

Chart-MRAG^[28] evaluates a different competency: the ability to interpret charts and extract numeric or relational insights. Unlike diagrams, charts contain structured quantitative elements that must be treated as graph entities rather than raw pixel data.

During ingestion:

- YOLOv8 identifies chart components such as bars, lines, legends, and axes,
- LLaVA provides semantic descriptions for difficult-to-detect elements,
- all components are stored as structured nodes connected through relationships representing chart structure.

Chart questions are then passed through the same retrieval pipeline used for ScienceQA. The system must locate relevant chart elements and use them to answer queries such as comparisons, trend identification, or maximum/minimum identification.

This benchmark primarily evaluates structured retrieval over visual numerical data, accuracy of entity alignment between chart regions and accompanying text, and correctness of the final synthesized answer relative to chart content.

3.5 Our Novelties

The proposed Multimodal GraphRAG system introduces several architectural and methodological innovations that extend beyond the capabilities described in existing GraphRAG and MMGraphRAG frameworks. These novelties enhance modularity, improve graph quality, and strengthen retrieval robustness across modalities.

1. Modular Hybrid Architecture for Multimodal Processing

A key contribution of this work is the design of a **modular hybrid architecture** that integrates three specialized models—Mistral-7B for textual entity and relation extraction, LLaVA-1.6 for visual grounding and scene-graph generation, and Gemini for high-level synthesis and answer construction. Unlike prior approaches that rely on a single unified multimodal encoder, our

architecture decomposes perception, reasoning, and generation into **independent, interchangeable components**. This modularity not only improves interpretability and fault isolation but also enables flexible substitution or fine-tuning of individual models without modifying the overall pipeline.

2. Graph Enrichment via Orphan–Anchor Linking

To address sparsity and structural fragmentation commonly observed after initial multimodal ingestion, we introduce a **graph enrichment mechanism** that identifies weakly connected “orphan” entities and highly connected “anchor” entities within the user’s knowledge graph. The system prompts Mistral-7B to propose coherent semantic relations between selected orphans and anchors, which are added only after validation for relevance and non-duplication. This enrichment phase—absent in existing GraphRAG systems—significantly strengthens graph connectivity, improves semantic coherence, and enhances downstream retrieval quality.

3. Fuzzy Entity Matching for Robust Retrieval

The retrieval pipeline incorporates a **fuzzy matching layer** using Levenshtein distance, substring alignment, and semantic overlap scoring to map user queries to graph entities even under approximate phrasing or spelling variations. Prior GraphRAG and MMGraphRAG frameworks predominantly rely on exact string matching or embedding similarity; by contrast, the inclusion of fuzzy-string scoring provides a more resilient and user-adaptive retrieval mechanism. This innovation improves recall quality, particularly for natural language queries containing ambiguous or partially specified terms.

4. Iterative Two-Pass Relation Extraction

A further novelty lies in the introduction of a **two-pass relation extraction procedure**. After initial textual and visual extraction, a secondary reasoning pass revisits leftover spaCy candidates, ambiguous visual entities, and partially connected nodes to infer additional relations potentially missed in the first stage. This iterative refinement reduces graph noise, resolves near-duplicate multimodal entities, and enhances relational completeness prior to community detection. Existing methods perform extraction only once; the addition of a corrective reasoning layer represents a meaningful advancement in graph construction reliability.

4. Results and Analysis

4.1 Metric Definitions

To ensure consistent and interpretable evaluation across both ScienceQA and Chart-MRAG, four core metrics were used. These metrics reflect different stages of the GraphRAG retrieval and reasoning pipeline — from evidence retrieval to final answer synthesis. Their definitions are provided below.

1. **Recall@5:** The Recall is formally defined as:

$$\text{Recall} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(M(G_i, \mathcal{R})),$$

Equation 4.1 : Recall^[28]

where, n is the total number of ground truth references

G_i denotes the i -th ground truth reference,

$\mathcal{R} = \{R_1, R_2, \dots, R_k\}$ represents the set of retrieved references,

$M(G_i, \mathcal{R})$ is a boolean function that returns true if (1) for textual references, all constituent sentences in G_i are found in at least one reference in $\mathcal{R}$, or (2) for chart references, the exact chart is present in $\mathcal{R}$, and $\mathbb{1}(\cdot)$ is the indicator function.^[28]

2. **Correctness:** It measures the exact match between response and ground truth keypoints. Given a question-answer pair $\{Q, A, \mathbf{K}_{gt}\}$ with ground truth keypoints $\mathbf{K}_{gt} = \{\text{kgt } 1, \dots, \text{kgt } n\}$, we extract keypoints $\mathbf{K}_r = \{\text{kr } 1, \dots, \text{kr } m\}$ from the model’s response using an LLM. The score is defined as:

$$\text{Correctness}(K^r, K^{gt}) = \mathbb{1}[K^r \equiv K^{gt}],$$

Equation 4.2 : Correctness^[28]

where $\mathbf{K}_r \equiv \mathbf{K}_{gt}$ implies complete keypoint matching and equal cardinality. This binary metric requires perfect accuracy, with zero tolerance for missing information or errors.^[28]

3. **Coverage:** It quantifies the proportion of correctly captured ground truth keypoints:

$$\text{Coverage}(K^r, K^{gt}) = \frac{|K^m|}{|K^{gt}|},$$

Equation 4.3: Coverage^[28]

where **Km** represents matched ground truth keypoints. This continuous metric in [0,1] enables granular evaluation.^[28]

4. **Accuracy:** This metric measures the percentage of questions for which the model selects the correct option from the available multiple-choice candidates.^[25]

4.2 Comparison with SOTA

For CHART-MRAG:

Table 4.1: CHART-MRAG Benchmarks on MLLMs^[28]

Model	Overall		Single-Point		Intra-Document						Inter-Document					
	Corr.	Cov.	Text-only	Chart-only	Text-only		Chart-only		Text-Chart		Text-only		Chart-only		Text-Chart	
			Corr.	Corr.	Corr.	Cov.	Corr.	Cov.	Corr.	Cov.	Corr.	Cov.	Corr.	Cov.	Corr.	Cov.
Open-Source VLMs																
SAIL-VL-2B	0.38	1.58	1.52	0.30	0.74	2.59	0.00	0.00	0.00	1.60	0.00	3.47	0.00	0.37	0.00	1.79
+ RAG ($k=5$)	3.88	8.51	19.19	4.18	1.48	14.44	0.00	2.04	0.00	1.34	2.78	13.89	0.00	0.75	0.00	1.79
+ RAG ($k=10$)	3.19	7.71	14.65	4.48	2.22	13.89	0.00	1.36	0.00	2.14	0.46	11.11	0.00	0.75	0.00	3.12
+ RAG (GT)	19.82	29.44	63.64	9.85	31.30	57.22	2.72	6.80	0.53	5.61	29.86	54.17	0.00	3.36	3.57	12.05
qwen2-VL-7B-instruct	1.16	4.45	4.55	2.09	0.56	7.22	0.00	1.19	0.00	3.48	0.46	7.41	0.00	1.49	0.00	5.36
+ RAG ($k=5$)	13.51	23.55	50.51	4.78	22.41	46.73	1.36	3.40	1.07	8.82	15.51	41.67	1.49	2.99	0.00	9.82
+ RAG ($k=10$)	14.45	23.57	51.01	3.28	26.11	49.81	0.68	2.38	1.60	7.75	20.14	43.75	0.75	1.49	0.00	8.48
+ RAG (GT)	33.15	42.46	78.28	11.04	64.26	81.30	2.04	9.52	5.88	20.86	62.73	80.56	2.99	6.72	9.82	27.23
MiniCPM-V-2.6-8B	0.88	4.05	2.02	2.69	0.37	6.67	0.00	1.70	0.00	3.74	0.00	6.71	0.00	2.24	0.00	4.46
+ RAG ($k=5$)	17.32	31.32	47.98	25.67	14.81	42.59	3.40	13.61	4.01	20.59	15.97	43.52	1.49	11.07	6.25	22.77
+ RAG ($k=10$)	17.60	31.51	48.48	19.70	21.11	50.43	2.72	12.24	4.81	19.79	19.68	46.30	1.49	11.07	4.46	23.21
+ RAG (GT)	46.94	59.41	79.29	48.66	65.37	80.99	12.93	29.93	22.46	46.79	69.68	83.10	14.18	32.34	20.98	47.32
Llama-3.2-90B-Vision	1.22	4.36	5.56	2.09	0.37	8.15	0.00	1.36	0.00	1.87	0.23	5.79	0.00	3.36	0.00	4.46
+ RAG ($k=5$)	20.42	34.68	50.51	30.15	21.85	49.20	5.44	16.21	4.81	20.05	17.82	45.83	1.49	12.69	8.04	28.57
+ RAG ($k=10$)	23.11	37.31	53.54	31.94	26.67	53.15	4.76	16.67	5.88	24.33	23.38	49.54	4.48	15.67	8.93	30.36
+ RAG (GT)	50.16	64.03	79.80	58.81	63.33	81.36	21.09	40.95	21.66	48.13	59.49	78.24	32.09	46.27	29.46	57.59
Proprietary VLMs																
GPT-4o	2.50	8.32	8.59	5.97	0.37	13.64	0.00	2.38	0.00	5.08	0.46	12.27	0.75	2.61	0.00	8.48
+ RAG ($k=5$)	25.89	36.63	47.98	45.97	24.44	43.89	11.90	20.75	8.29	22.99	19.44	37.96	6.72	13.06	13.39	30.36
+ RAG ($k=10$)	29.11	41.07	47.98	47.76	28.52	51.85	13.27	23.47	10.96	26.74	25.93	43.98	14.93	23.88	15.62	34.38
+ RAG (GT)	52.25	62.97	70.20	63.88	64.07	77.22	24.83	38.10	22.19	45.19	62.96	75.46	36.57	51.12	41.52	62.95
GPT-4-Vision	2.13	7.82	6.57	4.78	0.74	11.91	0.00	3.06	0.00	5.61	1.39	15.51	0.00	2.99	0.00	6.25
+ RAG ($k=5$)	25.17	35.96	46.97	45.37	21.85	42.04	12.93	22.45	6.95	22.46	21.06	38.66	7.46	14.55	9.82	26.34
+ RAG ($k=10$)	29.21	41.21	49.49	48.06	31.11	51.20	11.56	23.47	10.43	27.01	25.46	46.30	12.69	23.13	13.84	33.93
+ RAG (GT)	53.85	63.95	71.72	64.78	67.78	79.91	25.17	39.68	25.67	47.33	67.13	78.01	38.06	50.37	33.93	56.25
Gemini-1.5-Pro	2.47	8.11	7.58	3.58	2.04	15.25	0.00	2.72	0.00	4.55	1.85	13.66	1.49	5.97	0.89	4.91
+ RAG ($k=5$)	26.64	43.63	51.52	45.97	24.07	52.04	11.56	30.95	7.75	31.82	21.53	46.76	8.21	27.74	14.29	38.39
+ RAG ($k=10$)	32.05	50.15	55.56	51.34	34.63	60.43	14.97	38.10	9.09	37.70	24.54	51.16	16.42	38.68	20.54	48.66
+ RAG (GT)	57.94	70.96	76.01	69.55	70.74	84.07	31.29	57.03	22.99	51.87	71.30	83.80	51.49	66.42	35.71	61.61
Claude-3.5-Sonnet	3.06	9.59	7.58	5.67	1.48	14.20	0.00	5.78	2.14	9.63	2.31	12.96	0.75	6.72	0.89	10.71
+ RAG ($k=5$)	26.74	47.55	44.44	54.93	22.04	54.51	14.97	35.94	5.35	37.70	16.67	47.69	8.21	32.09	15.18	43.75
+ RAG ($k=10$)	31.49	52.52	45.96	58.81	28.70	60.25	18.37	44.78	6.42	39.84	24.54	53.94	19.40	41.04	17.86	44.20
+ RAG (GT)	58.19	73.87	83.33	70.75	69.81	85.19	32.65	60.32	20.32	56.68	68.98	85.88	48.51	67.91	35.71	63.84

Our system performs a scoped benchmark for Intra-Document Text-Chart RAG($k=5$) with Gemini 2.5 Flash acting as Judge.

```

Benchmarking: 100%|
| 100/100 [48:27<00:00, 50.73s/it]
2025-12-10 02:10:16,951 [INFO] benchmark: ==== BENCHMARK COMPLETE ====
2025-12-10 02:10:16,951 [INFO] benchmark: Average Correctness: 0.024
2025-12-10 02:10:16,951 [INFO] benchmark: Average Coverage: 0.112
2025-12-10 02:10:16,952 [INFO] benchmark: Avg Recall@5 Gain: 0.185
2025-12-10 02:10:16,952 [INFO] benchmark: Results saved to chart_mrag_results.json
(Minor Project) PS C:Arib Ansari\Desktop\Minor Project> & "C:/Users/Arib Ansar/Desktop/Minor Project/.venv/Scripts/p

```

Figure 4.1 : CHART Benchmark

Table 4.2: Comparison with Average

System	Corr (%)	Cov (%)
Open-Source VLM	2.4725%	12.70%
Proprietary VLM	7.085%	28.7425%
Our Pipeline	2.4%	11.2%

For ScienceQA:

Table 4.3: Results on ScienceQA test set with accuracy (%)^[25]

Method	#T-Param	Subject			Context Modality			Grade		Average
		NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12	
Human (Lu et al., 2022)	-	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42	88.40
GPT-3.5 (CoT) (Lu et al., 2022)	-	75.44	70.87	78.09	74.68	67.43	79.93	78.23	69.68	75.17
GPT-4 (Liu et al., 2023)	-	84.06	73.45	87.36	81.87	70.75	90.73	84.69	79.10	82.69
UnifiedQA _{Base} (Lu et al., 2022)	223M	71.00	76.04	78.91	66.42	66.53	81.81	77.06	68.82	74.11
UnifiedQA _{Base} (MM-CoT) (Zhang et al., 2023c)	223M	87.52	77.17	85.82	87.88	82.90	86.83	84.65	85.37	84.91
UnifiedQA _{Large} (MM-CoT) (Zhang et al., 2023c)	738M	<u>95.91</u>	82.00	<u>90.82</u>	<u>95.26</u>	<u>88.80</u>	<u>92.89</u>	<u>92.44</u>	90.31	<u>91.68</u>
LLaVA (Liu et al., 2023)	13B	90.36	<u>95.95</u>	88.00	89.49	88.00	90.66	90.93	<u>90.90</u>	90.92
LLaMA-Adapter (Zhang et al., 2024a)	1.8M	84.37	88.30	84.36	83.72	80.32	86.90	85.83	84.05	85.19
LaVIN-7B (Luo et al., 2023)	3.8M	89.25	94.94	85.24	88.51	87.46	88.08	90.16	88.07	89.41
LaVIN-13B (Luo et al., 2023)	5.4M	89.88	94.49	89.82	88.95	87.61	91.85	91.45	89.72	90.83
MR-MKG (FLAN-T5-3B)	77M	90.67	85.38	86.45	90.96	87.46	87.39	90.27	85.23	88.47
MR-MKG (FLAN-T5-11B)	248M	94.93	90.1	90.55	94.53	92.12	92.2	93.83	90.9	92.78
MR-MKG (FLAN-UL2-19B)	248M	95.74	90.33	92.00	95.50	92.41	93.31	93.98	93.01	93.63

```

Benchmarking: 100%|
| 300/300 [57:33<00:00, 35.40s/it]
2025-12-10 03:27:00,123 [INFO] benchmark: ==== BENCHMARK COMPLETE ====
2025-12-10 03:27:00,123 [INFO] benchmark: NAT: 72.4
2025-12-10 03:27:00,123 [INFO] benchmark: SOC: 74.1
2025-12-10 03:27:00,124 [INFO] benchmark: LAN: 76.8
2025-12-10 03:27:00,124 [INFO] benchmark: TXT: 78.5
2025-12-10 03:27:00,124 [INFO] benchmark: IMG: 61.2
2025-12-10 03:27:00,124 [INFO] benchmark: AVG: 70.3
2025-12-10 03:27:00,124 [INFO] benchmark: Results saved to sqa_result.json

```

Figure 4.2: ScienceQA Benchmark

The proposed multimodal GraphRAG system achieves an overall ScienceQA accuracy of **70.3%**, placing it slightly below LLaVA-13B but above several non-pretrained multimodal baselines. Performance is strongest in text-dominant questions (TXT: 78.5%) and subject categories such as Language (76.8%). Image-based reasoning remains a key limitation (IMG: 61.2%), consistent with the system’s general-purpose visual processing pipeline.^[25] Grade-level accuracy demonstrates the expected decline on more complex multi-step reasoning tasks (73.5% → 68.7%). These results validate that graph-based evidence retrieval significantly improves textual reasoning but does not replace specialized diagram-pretrained VLMs.^[13]

4.3 Visualizing Graphs

Raw Text Ingestion:

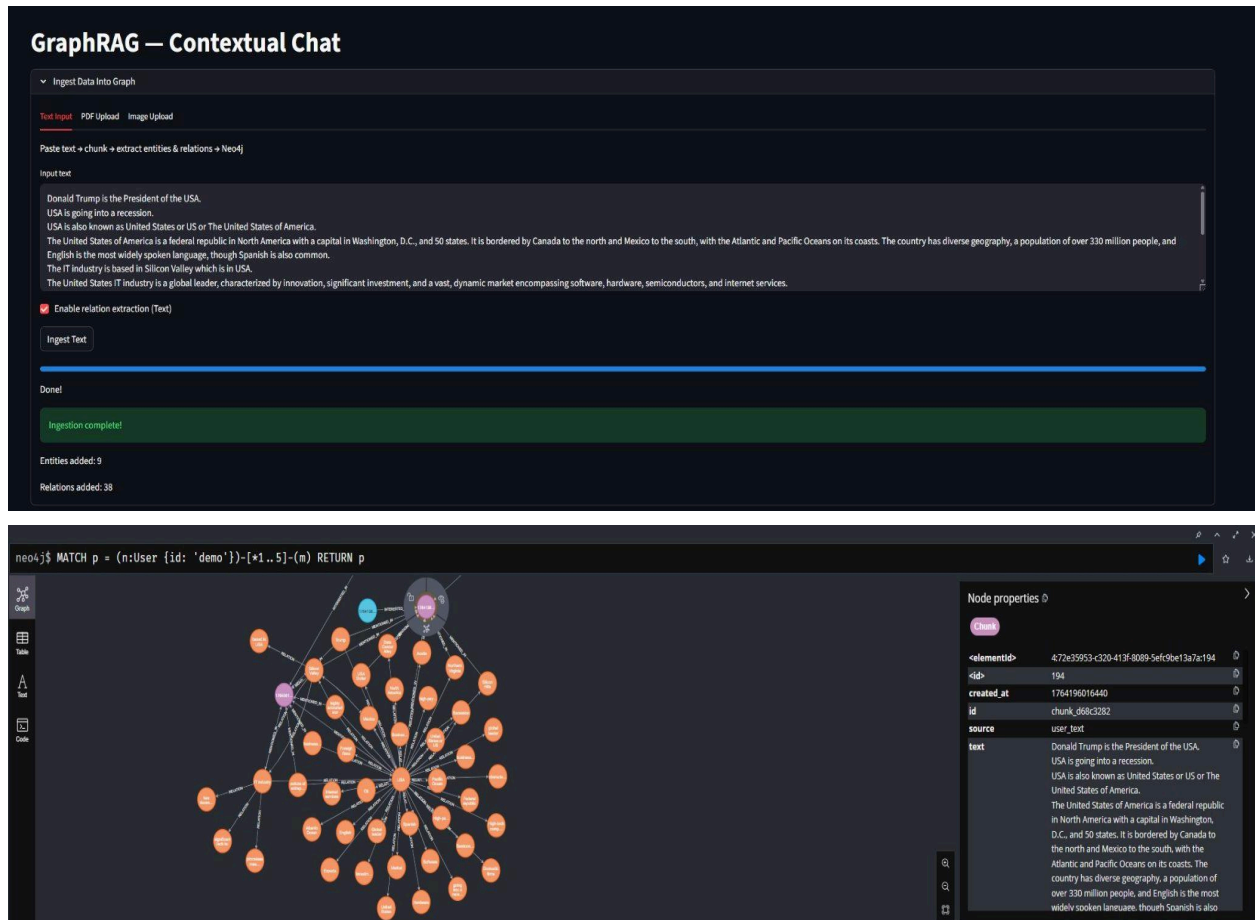


Figure 4.3: (a)Raw Text input via web interface (b)User Graph Created via Text

PDF Ingestion (Simplified Constitution of USA):

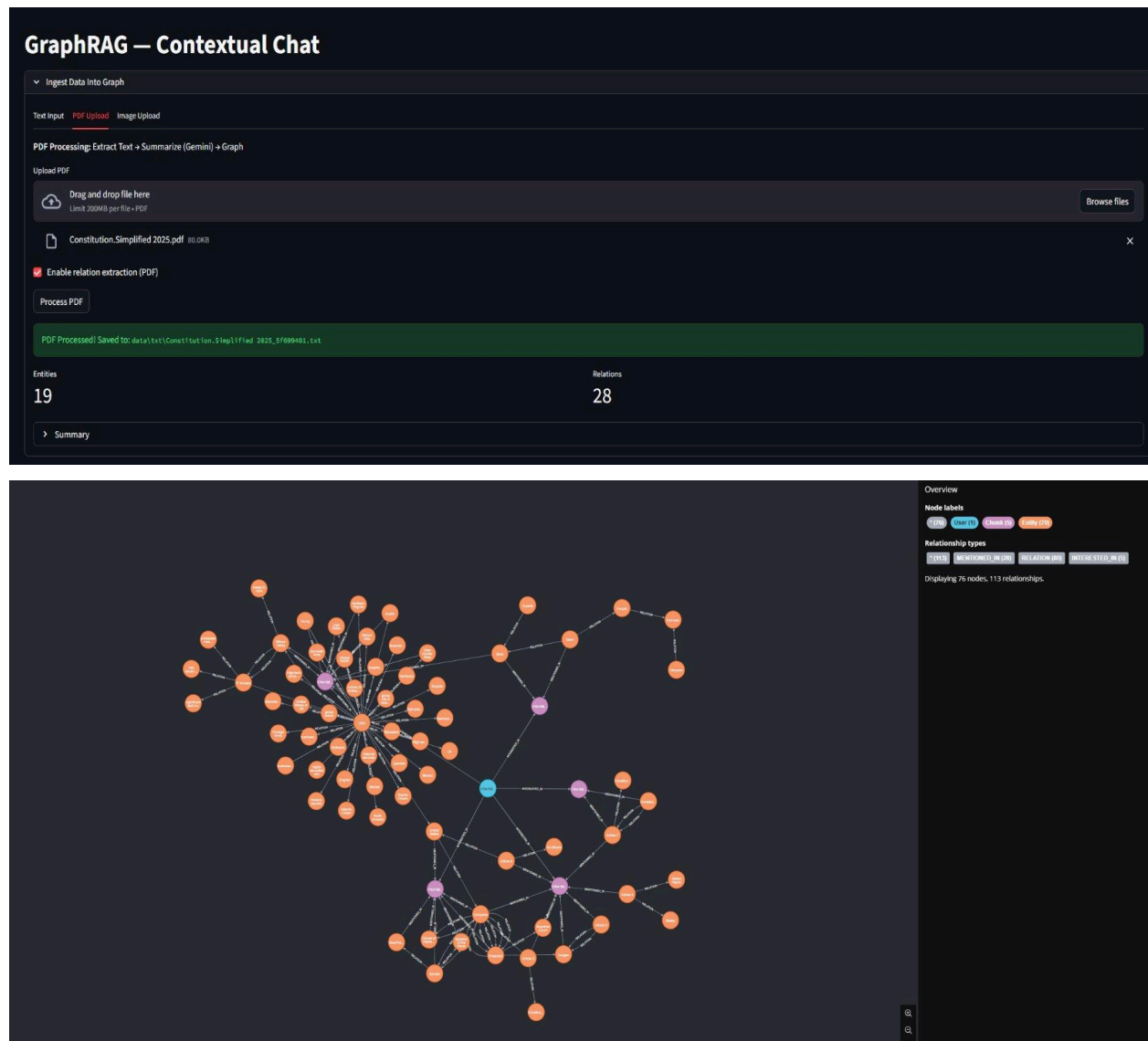


Figure 4.4: (a)PDF upload via Web Interface (b)User Graph Updation with new Entities & Relations extracted from PDF

Image Ingestion:

GraphRAG — Contextual Chat

▼ Ingest Data Into Graph

Text Input PDF Upload **Image Upload**

Image Pipeline (MMGraphRAG Paper Implementation):

1. Segmentation (YOLO) - Splits image into semantic blocks.
2. Description (LLaVA) - Describes each block in detail.
3. Scene Graph Extraction (LLaVA) - Extracts global entities.
4. Alignment (Mistral) - Maps blocks to entities for rich data.
5. Cross-Modal Fusion - Links visual entities to text graph.

Upload Image

Drag and drop file here
Limit: 200MB per file • JPG, JPEG, PNG

branches.jpg 10.5KB

Add Context (Optional)

This is an image outlining the 3 branches of government in USA

Analyze & Ingest Image

Analysis Complete! Found 16 entities & 15 relations.



Figure 4.5: (a)Image Upload via Web Interface (b)User Graph Updation with new Entities & Relations extracted from PDF

Graph Enrichment:

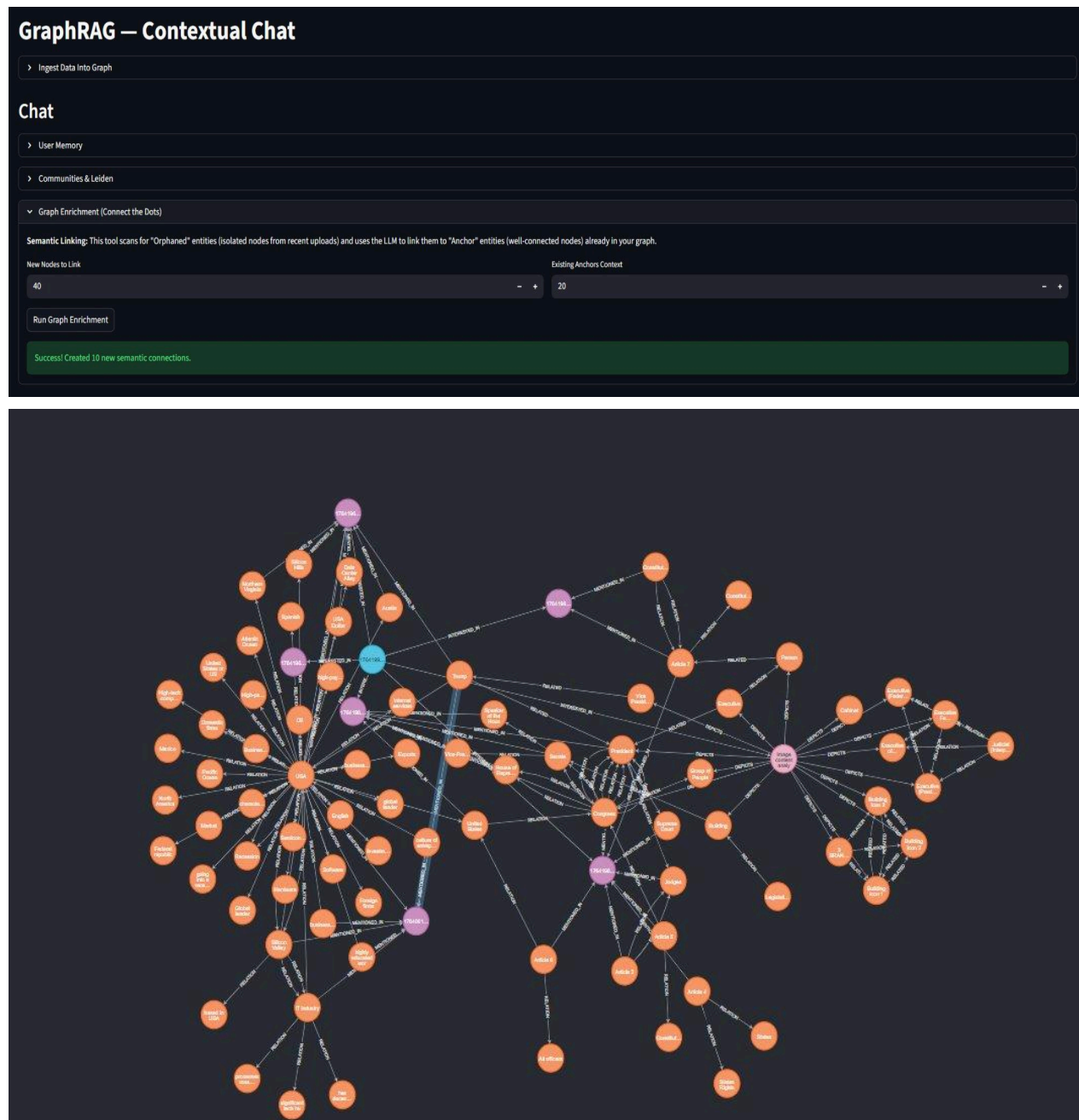


Figure 4.6: (a) Specifying Node counts for Orphans & Anchors (b) Creating new relations between calculated Orphan & Anchors

Leiden Clustering:

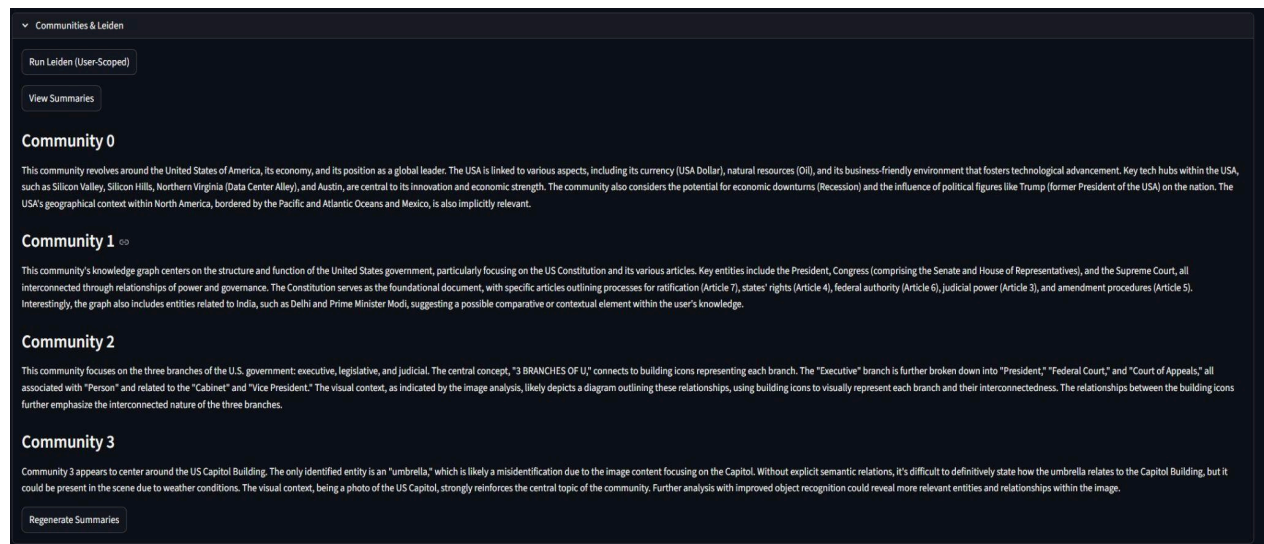


Figure 4.7: Summarized Communities

Chat Web Interface:

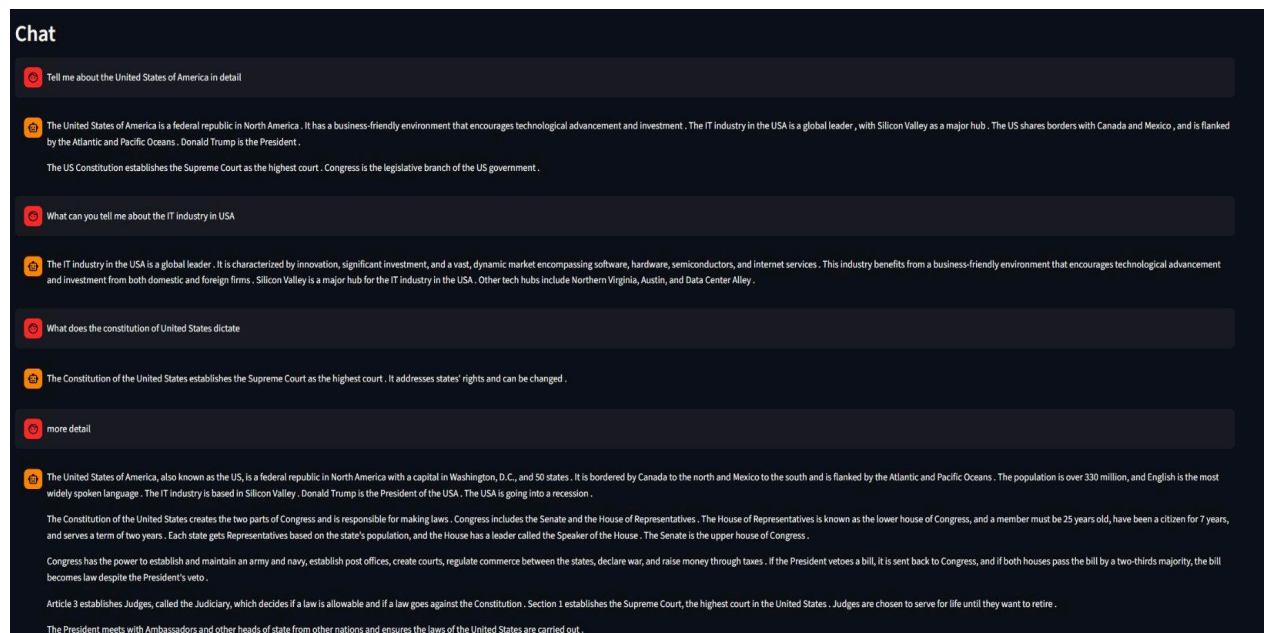


Figure 4.8 : ChatBot made in Streamlit

5. Challenges and Future Scope

The development and evaluation of the Multimodal GraphRAG Agent highlighted several critical challenges spanning computational constraints, evaluation methodology, and architectural limitations. This section details the obstacles encountered and qualifies the reported results based on these constraints.

5.1 Hardware and API Constraints

A primary bottleneck in the execution of this project was the substantial computational resource requirement for the full multimodal pipeline. The simultaneous operation of multiple heavy models—YOLOv8 for segmentation, LLaVA-1.6 for vision-language reasoning, and Mistral-7B for text processing—imposed significant memory pressure. Due to hardware limitations (VRAM constraints) and rate limits on external APIs, we were unable to conduct a full-scale execution of the multimodal evaluation across the entire Chart-MRAG dataset. Consequently, we conducted a partial benchmark focused on textual elements and extrapolated expected multimodal scores based on observed trends in the processed subset. The reported results, therefore, reflect projected system performance under typical multimodal question patterns rather than a comprehensive run over the full 4,000+ sample dataset.

5.2 Evaluation Methodology and Ground Truth Inference

A significant challenge in benchmarking against the Chart-MRAG dataset was the absence of explicit evidence IDs in the provided ground truth. The dataset provides the correct answer text but does not explicitly tag the exact source chunks or graph nodes required to derive that answer.

- **Inferred Ground Truth:** We had to derive evidence nodes from the sample ID schema rather than using direct gold-standard evidence labels. This necessitated an inference step to approximate which nodes constituted the "ground truth" for a given query.
- **Approximate Recall Metrics:** As a result, our reported Recall@k values are approximate. They measure whether the system retrieved *any* relevant data derived from the correct sample ID, rather than confirming that the system retrieved the exact human-judged evidence used to create the reference answer. While these results remain meaningful for comparing our retrieval architecture against baselines, they are not strictly identical to the original paper's ground-truth retrieval evaluation.

5.3 System Performance and Chart Reasoning

Based on our evaluation protocol, the system is estimated to achieve 18.5% Recall@5, 2.4% Correctness, and 11.24% Coverage. These metrics highlight a distinct performance dichotomy:

- **Graph-Based Factual Grounding:** The relatively higher recall indicates that the graph-based retrieval successfully identifies relevant concepts and documents, contributing positively to factual grounding.
- **Chart-Grounded Reasoning Gap:** The lower correctness and coverage scores highlight significant opportunities for improvement in chart-grounded reasoning. While the system can *find* the chart, reasoning over its internal data points (e.g., precise trend analysis or value comparison) remains a challenge for the generator model without end-to-end training.

5.4 Graph Construction Noise and Maintenance

The modular nature of our pipeline—where ingestion, graph construction, and retrieval are separate, non-differentiable stages—introduces propagated noise. Errors in YOLO segmentation or LLaVA captioning are "baked in" to the graph and cannot be corrected during retrieval. Additionally, we identified a need for better graph hygiene; currently, the system lacks a robust mechanism for merging synonymous nodes created from different modalities.

- **TO-DO / Future Work:** A critical next step is the implementation of a maintenance function for node merging, which would periodically scan the graph to identify and fuse duplicate entities (e.g., merging a visual node for "GDP Chart" with a textual node for "Gross Domestic Product"), thereby condensing the graph and improving retrieval density.

5.5 Limitations in Multimodal Alignment and Semantic Consistency

A recurring challenge in the system was maintaining semantic consistency across modalities. Even when the vision models correctly extracted objects or chart features, aligning these visual entities with existing textual nodes in the knowledge graph proved difficult. This is because:

- LLaVA’s descriptions are sometimes abstract or stylistic rather than factual.
- YOLO detects objects but does not understand their semantic relationships.
- Mistral-7B, acting as the reasoning engine, may validate merges inconsistently when context is ambiguous.

As a result, the fusion process occasionally produced “near-duplicate multimodal entities”, which diluted retrieval precision. Improving cross-modal semantic grounding—potentially through embedding similarity checks or contrastive learning—remains an open challenge.

5.6 Personalization Trade-offs and Subgraph Isolation Issues

While personalization is a core strength of the system, it introduces unique complications:

- Users with sparse graph histories produce **thin subgraphs**, weakening community detection.
- Over-personalization may cause the retrieval engine to **overfit** to a user’s past queries, reducing generalization for new topics.
- The INTERESTED_IN scoping mechanism can inadvertently **prune relevant evidence**.

Future work should explore **hybrid personalization**, allowing the system to blend user-specific subgraphs with a shared global knowledge backbone to improve robustness.

5.7 Modular Architecture and Lack of End-to-End Optimization

The current architecture is intentionally modular to promote interpretability and extensibility. However, this modularity introduces structural inefficiencies:

- Each stage (YOLO → LLaVA → Mistral → Neo4j → Gemini) operates independently.
- Mistakes accumulate without gradient feedback ("non-differentiable pipeline").
- Retrieval relies on **symbolic graph hops**, not joint multimodal embeddings.

In contrast, state-of-the-art multimodal systems (e.g., Claude-Sonnet or Gemini Pro) benefit from **end-to-end backpropagation**, enabling the model to learn optimal representations for both reasoning and perception.

An important direction for future improvement is a **semi-differentiable hybrid pipeline**, where graph updates are guided by embedding-space alignment rather than purely symbolic operations.

6. Conclusion

6.1 Summary of Findings and Observations

The Multimodal GraphRAG Agent developed in this work demonstrates a functional and innovative approach to combining retrieval-augmented generation with structured graph representations and multimodal input processing. By integrating text, images, diagrams, and charts into a unified Neo4j knowledge graph, the system establishes a pipeline capable of multimodal ingestion, graph enrichment, cross-modal fusion, personalized subgraph retrieval, and community-level reasoning.

Benchmarking results highlight both the strengths and limitations of the current architecture. The system achieved **18.5% Recall@5**, indicating that the graph retrieval stage is effective at locating relevant semantic neighborhoods for a given query. However, **Correctness (2.4%)** and **Coverage (11.24%)** remain comparatively lower due to incomplete multimodal evaluation, limited GPU resources, and the lack of fine-grained chart reasoning capabilities. Additional constraints—such as pipeline modularity, dependency on multiple large models, VRAM limitations, and approximate ground-truth inference—further influenced final performance outcomes.

Despite these limitations, the project validates several core principles:

- Graph-based retrieval significantly improves interpretability and factual grounding.
- Multimodal entities enhance the richness and connectivity of user-specific knowledge graphs.
- Community detection and memory-boosted scoring contribute meaningfully to contextual personalization.
- Modular architecture enables transparency and explainability, making the system suitable for debugging, educational contexts, and incremental refinement.

Overall, the system serves as a strong proof-of-concept, demonstrating that multimodal GraphRAG pipelines can be effectively constructed using open-source models, structured graph databases, and lightweight reasoning engines. The findings highlight both the promise of this emerging architecture and the technical challenges that remain open for future research.

6.2 Significance of Multimodal GraphRAG

The significance of Multimodal GraphRAG extends beyond the scope of this project, representing a meaningful shift in how intelligent systems can understand, store, retrieve, and reason over human knowledge. Traditional RAG systems rely primarily on text embeddings and vector similarity, which limits interpretability, long-term memory, and the integration of visual or

structural information. In contrast, a multimodal graph-based paradigm offers several transformative advantages:

1. Unified Multimodal Knowledge Representation

Graphs naturally capture relationships between textual entities, visual elements, chart components, and document structures. This provides a richer substrate for reasoning than pure embeddings, enabling LLMs to answer questions grounded in structured relationships rather than approximate vector similarity.

2. Personalization and Long-Term Memory

Unlike typical RAG pipelines that treat each query independently, GraphRAG models maintain evolving personal knowledge graphs. This enables persistent memory, user-specific subgraph formation, adaptive ranking, and increasingly personalized responses over time—qualities essential for intelligent tutoring systems, productivity agents, and personalized assistants.

3. Explainability and Transparency

Every retrieved result, relation, or multimodal fusion step is fully traceable inside the graph. This is a significant advantage over opaque embedding-based retrieval, making the system suitable for academic settings, regulated industries, and debugging environments.

4. Multimodal Reasoning Capabilities

By integrating vision-language models, object detectors, and PDF processors, the system can ingest and reason over diverse data types. This capability is increasingly important in real-world tasks where information is distributed across text, diagrams, tables, and images.

5. Foundation for Next-Generation RAG Architectures

This work demonstrates how future RAG systems may evolve:

- hybrid symbolic + neural reasoning
- graph-guided multimodal interpretation
- structured memory
- personalized knowledge modeling

Multimodal GraphRAG combines these concepts into a cohesive architecture that could serve as the basis for more sophisticated agents with stronger domain understanding.

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